# Detailed PDC Target-Momentum Reconstruction and Uncertainty Analysis

## Methods, Error Models, and Confidence/Credible Intervals

SMSimulator5.5 Analysis

April 11, 2026

# Study Scope

**Physics question**

- Reconstruct proton target momentum from a two-point PDC track in the SAMURAI field.
- Quantify not only the central estimate, but also the uncertainty attached to that estimate.

**This report focuses on**

- Runtime reconstruction backends now exposed by `reconstruct_target_momentum`
- RK-based uncertainty analysis implemented in `analysis_pdc_reco`
- Diagnostic comparison between RK, NN, and multidimensional regression branches
- Confidence and credible intervals for $p_x$, $p_y$, $p_z$, and $|\vec{p}|$

# Top-Level Reconstruction Methods

| Method | Input / model | Current strength | Current limitation |
|---|---|---|---|
| RK fit | Field-map back-tracing from target to PDC constraints | Physics-based, interpretable, supports uncertainty propagation | Slowest branch; quality depends on initialization and fit mode |
| NN regressor | Direct map from PDC coordinates to $(p_x, p_y, p_z)$ | Best point-estimate accuracy on current truth grid | No interval output in runtime yet |
| MultiDim surrogate | Polynomial surrogate from offline regression | Fast when calibrated | Current truth-grid sample has no matched events; no interval support |

The current uncertainty study is therefore centered on the RK branch, because it is the only branch that now exposes Fisher, profile-likelihood, Laplace, and MCMC interval estimates.

# RK Reconstruction Modes

**The RK backend currently supports three fit modes**

1. **two-point-backprop**: infer momentum by back-propagating the PDC line to the target in the field map.
2. **fixed-target-pdc-only**: keep the target anchor fixed and fit the momentum state against the two PDC points.
3. **three-point-free**: fit both target point and momentum state as free parameters under target and PDC constraints.

**Why this matters for uncertainty**

- The parameterization determines the Hessian conditioning and therefore the Fisher covariance.
- The fit mode also changes the profile-likelihood geometry and the posterior shape.
- In the present implementation, the enlarged legacy-run analysis uses `fixed-target-pdc-only`, because it is the most stable mode for real-style reconstructed input.

# What Is Written by the RK Error Analysis

**Runtime ROOT branches now available**

- Central fit diagnostics: status, method, iterations, raw and reduced $\chi^2$, $N_{\mathrm{dof}}$
- Fisher widths: $\sigma_{p_x}$, $\sigma_{p_y}$, $\sigma_{p_z}$, $\sigma_{|\vec{p}|}$
- Optional Laplace intervals: 68% and 95% lower/upper bounds for $p_x$, $p_y$, $p_z$, and $|\vec{p}|$

**Offline diagnostic tool**

- `build/bin/analyze_pdc_rk_error`
- Outputs: `track_summary.csv`, `profile_samples.csv`, `bayesian_samples.csv`, `summary.csv`, `validation_summary.csv`, and summary plots
- This tool re-runs the RK fit on reconstructed PDC tracks and computes frequentist and Bayesian intervals on sampled candidates

# Interval Definitions

| Estimate | Construction | Statistical meaning | Current use |
|---|---|---|---|
| Fisher | Inverse local Hessian around the RK optimum | Local Gaussian approximation to the likelihood | Fast width estimate for all successful tracks |
| Profile likelihood | Scan one fit parameter, minimize others, use $\Delta\chi^2 = 1, 4$ crossings | Frequentist confidence interval | Asymmetry and nonlinearity diagnosis on sampled tracks |
| Laplace posterior | Gaussian posterior near the mode using posterior covariance | Bayesian credible interval | Fast Bayesian interval for all tracks if posterior is valid |
| MCMC posterior | Random-walk samples from the posterior | Bayesian credible interval with shape diagnostics | Validation on sampled tracks, not on every event |

**Important distinction**: Fisher and Laplace are local approximations; profile likelihood and MCMC are more robust to non-Gaussian structure but much slower.

# Synthetic Truth-Grid Method Comparison

**Diagnostic dataset**

- Existing five-point truth grid around $p_z = 627$ MeV/c
- One matched event per truth point in the current regression fixture
- Use this only as a backend sanity check, not as a final performance benchmark

| Backend | Matched events | Mean $\lvert\mathrm{bias}_{p_x}\rvert$ | Mean $\lvert\mathrm{bias}_{p_y}\rvert$ | Mean $\lvert\mathrm{bias}_{p_z}\rvert$ |
|---|---|---|---|---|
| NN | 5 | 21.04 | 15.56 | 3.10 |
| RK fixed-target | 5 | 178.35 | 11.46 | 234.39 |
| RK three-point | 5 | 651.35 | 15.77 | 190.31 |
| RK two-point | 5 | 67.99 | 1.40 | 604.92 |
| MultiDim | 0 | n/a | n/a | n/a |

On this fixture, the NN branch is currently the strongest point estimator, while the RK modes still need calibration or configuration tuning for this truth-grid operating point.

# How to Read the Truth-Grid Result

## Interpretation

- The truth-grid test is very small, so the table is dominated by pointwise behaviour rather than stable statistics.
- Even so, the pattern is useful: each RK mode fails differently.
- `two-point-backprop` keeps transverse components relatively close, but collapses $p_z$.
- `fixed-target-pdc-only` is more balanced and therefore chosen as the first uncertainty-analysis mode for legacy reconstructed samples.
- `three-point-free` is the most flexible and also the least stable on the current small truth-grid fixture.

## Practical conclusion

- Use the NN branch when the immediate goal is best point prediction.
- Use the RK branch when the goal is uncertainty-aware reconstruction and physically interpretable interval estimates.

# Expanded Legacy Reconstruction Sample

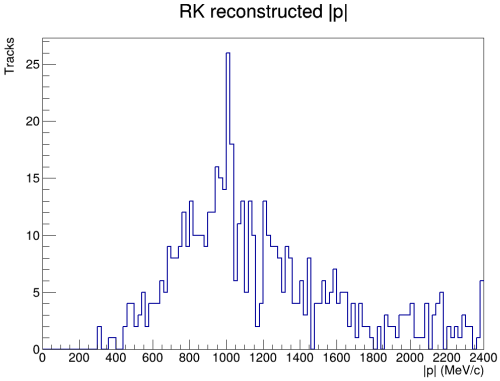
**Input class**

- Legacy reconstructed `after_pdcana` ROOT files
- Geometry macro: `geometry_B115T_pdcOptimized_20260227.mac`
- Magnetic field map: `180703-1,15T-3000.table`, dipole rotation 30°
- RK mode: `fixed-target-pdc-only`

**Expanded run configuration**

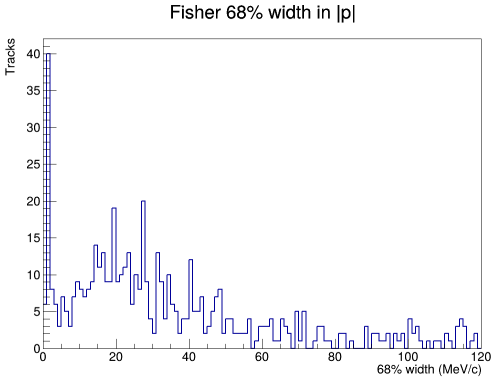
- Up to 500 events per file
- Profile scan: 11 points per parameter, 4 sampled tracks per momentum quartile
- Bayesian validation: 2 sampled tracks per quartile, 120 kept MCMC samples after burn-in/thinning
- Current completed output: 815 successful RK tracks, 32 profile samples, 16 Bayesian samples

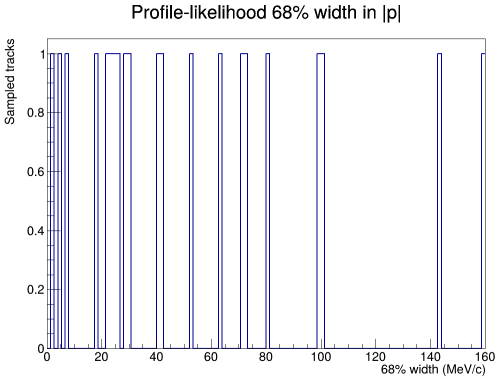
# Reconstructed Momentum Spectrum




The enlarged legacy sample covers a broad reconstructed proton momentum range. The median $|\vec{p}|$ is 1370.4 MeV/c with interquartile range 970.8 to 2920.8 MeV/c.

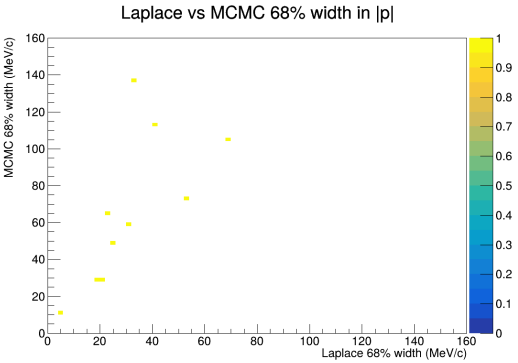
# Fisher and Profile-Likelihood Widths




In the enlarged run, the median 68% width in $|\vec{p}|$ is 48.20 MeV/c from the Fisher estimate and 76.91 MeV/c from profile likelihood. The profile interval remains substantially broader, so the likelihood is still not well described by a purely local Gaussian approximation.

# Laplace Versus MCMC



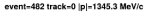

**Enlarged run**

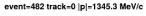
- Median Laplace 68% width in $|\vec{p}|$: 48.14 MeV/c
- Median MCMC 68% width in $|\vec{p}|$: 109.32 MeV/c
- Median MCMC acceptance rate: 0.593
- Median effective sample size: 9.23

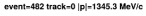
**Reading**

- Laplace tracks Fisher closely, as expected for two local Gaussian approximations.
- MCMC is much broader on sampled tracks, which again suggests non-Gaussian posterior structure.

# Representative Profile Curves




This is the clearest frequentist diagnostic. Symmetric parabolic curves would support Fisher-like treatment. Asymmetry, flat directions, or clipped crossings explain why profile-likelihood intervals can be wider than Hessian-based widths.

# Confidence and Credible Interval Summary

| Method | Tracks covered now | 68% interval type | Main message |
|---|---|---|---|
| Fisher | All successful RK tracks | Local confidence width | Good for speed, optimistic when the likelihood is curved or asymmetric |
| Laplace | All successful RK tracks with valid posterior covariance | Local Bayesian credible interval | Similar to Fisher in the current smoke study |
| Profile likelihood | Sampled RK tracks | Frequentist confidence interval | Captures asymmetry and broader nonlocal structure |
| MCMC | Sampled RK tracks | Bayesian credible interval | Best shape information, but highest runtime cost |

For current PDC legacy reco data, the safest interpretation is: Fisher/Laplace for fast per-track monitoring, profile/MCMC for method validation and interval realism checks.

# Current Limitation on Coverage Validation

**What is missing**

- The legacy `after_pdcana` files contain an `OriginalParticleData` branch without a safe dictionary in this workflow.
- Direct truth binding from those files caused the earlier memory-corruption crash and has been disabled in the analysis tool.

**Consequence**

- We can compute interval sizes and compare methods on reconstructed tracks.
- We cannot yet claim empirical 68% or 95% coverage on the legacy sample.
- Coverage validation must be done on simulation files with a safe truth interface or by adding a reliable dictionary-backed legacy reader.

# Implementation Status

**Completed**

- RK runtime branches for central fit diagnostics and interval outputs
- Offline tool for Fisher, profile-likelihood, Laplace, and MCMC comparison
- Synthetic backend comparison CSV summaries
- Stability fix for the legacy-file crash path caused by unsafe truth-branch binding

**Still needed**

- Calibrate RK modes against larger truth-labeled samples
- Add interval support for the NN branch, for example heteroscedastic regression or quantile targets
- Restore truth-based coverage validation for legacy-style files through a safe reader path

# Takeaway

- The PDC reconstruction framework now supports several central-estimate methods, but only the RK branch currently supports rigorous uncertainty analysis.
- On the available legacy reconstructed sample, local Gaussian intervals are much narrower than profile-likelihood and MCMC intervals, so Gaussian-only error bars are not sufficient.
- The NN branch remains the strongest point predictor on the current truth-grid diagnostic, but it is not yet uncertainty-aware.
- The next technically important step is not another point-estimate tweak; it is truth-backed interval validation on a larger, stable dataset.